Service Utilization Analytics

Historical service-demand reporting using 20,190 public RAND study records

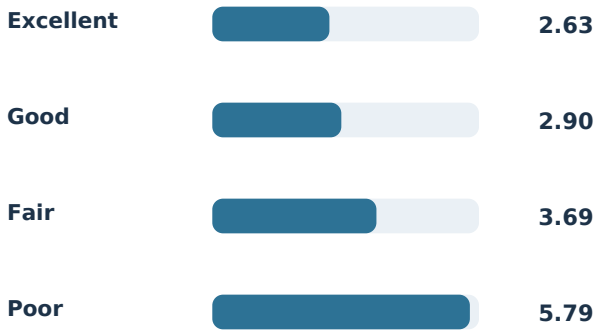
EXECUTIVE SNAPSHOT

Question: How are outpatient physician visits distributed across study records, and what do the health-status and coinsurance segments reveal?

20,190	57,752	2.86	31.2%
STUDY RECORDS	MD VISITS	MEAN / RECORD	ZERO VISITS

Visits by health status

Average outpatient physician visits



Zero visits by cost share

% of study rows with no MD visits

THREE TAKEAWAYS

01 / Skewed demand. 6,308 rows report no physician visit. Mean visits alone hides the long tail of demand.

02 / Variation by health. “Poor” health records average 5.79 visits (n=302), versus 2.63 for “Excellent” (n=11,019).

03 / Concentrated activity. Just 9.2% of records (8+ visits) account for 42.6% of all recorded visits.

HOW TO INTERPRET THIS

This is historical research data, not current patient data or a staffing forecast. A record is not a unique person. Patterns are descriptive, not standalone causal estimates.

An auditable analytics workflow

Python transformations, SQL checks, spreadsheet reporting, and an interactive web dashboard

WORKFLOW

01
SOURCE
statsmodels RAND HIE

02
QA
validate all fields

03
MODEL
define clean measures

04
REPORT
SQL / Excel / web

Checks performed

Results independently verified in SQL

Records and row keys	**20,190 / 20,190 unique**
Critical missing values	**0**
Negative visit counts	**0**
Invalid cost-share groups	**0**
Inconsistent health flags	**0**
Python vs SQL visit total	**MATCH**

Analytical choices

Exact definitions prevent false claims

Visits: count of outpatient MD visits per source study record.

Health: derived from good/fair/poor flags; “Excellent” is the omitted category.

Coinsurance: percentage reconstructed as round(exp(lncoins) − 1).

Risk: groups are unbalanced; do not treat unadjusted gaps as cause and effect.

SQL | A REPRODUCIBLE METRIC CHECK

```
SELECT COUNT(*) AS records,
       SUM(outpatient_visits) AS total_visits,
       ROUND(AVG(outpatient_visits), 2) AS mean_visits,
       SUM(CASE WHEN outpatient_visits=0 THEN 1 ELSE 0 END) AS zeros
FROM visits;
```

PORTFOLIO PACKAGE

Includes interactive index.html, runnable Python, SQLite SQL views, data CSV, and a formula-based Excel workbook. Source: RAND Health Insurance Experiment subset (distributed by statsmodels; documented as public domain).

Important: This is a public-data demonstration for portfolio review, not an employment project. The owner should inspect the code and verify the narrative before publishing.